Yaashi Mallik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.93
Narayana Junior College	Class XII (Telangana State Board)	2024	96.40%
Meridian International School	Class X (CBSE)	2022	97.20%

PROJECTS

Pathfinding Visualizer | React, C++, REST API

- **Engineered a full-stack pathfinding visualizer** by decoupling an interactive React frontend from a high-performance C++17 REST API backend to handle computationally intensive graph traversals efficiently. Configured custom HTTP POST endpoints using **cpp-httplib** and **nlohmann/json** to serialize, transmit, and parse user-defined obstacle grids and coordinates with minimal latency.
- **Implemented Dijkstra's and A* (A-Star) graph traversal algorithms** in C++ to compute optimal shortest paths across 2D grids, leveraging Manhattan distance heuristics in A* to significantly prune search space exploration.
- **Modeled the grid as an implicit graph** with 4-directional connectivity, utilizing priority queues (min-heap) for efficient node relaxation, visited matrices for O(1) cycle detection, and parent-pointer matrices to reconstruct paths via backtracking.

Custom Semantic Search Engine | Python, Docker, Qdrant, OpenAI API

- Built a GenAI-powered semantic search tool -- scraping, chunking, and embedding web content into a vector database for meaning-based retrieval, the same architecture behind enterprise AI copilots and knowledge-search tools.
- Deployed a production-style vector database (Qdrant, Docker), demonstrating hands-on ability to build and operate AI infrastructure, not just call an API.
- Designed a reusable scraper -> embedder -> vector DB pipeline, applicable to business use cases like internal document search or support automation.

Market Basket Analysis | Python, Pandas, MLxtend

- Conducted a quantitative market basket analysis on 9,835 retail transactions, applying the Apriori algorithm to mine frequent itemsets and generate association rules at scale.
- Ranked rules by support, confidence, and lift to isolate statistically significant product affinities (lift > 2), distinguishing genuine cross-sell signal from noise.
- Translated the analysis into a data-backed bundling and shelf-placement strategy, the kind of quantitative groundwork used in retail/CPG consulting engagements to size cross-sell revenue opportunities.

SKILLS

Programming Languages: Python, C, C++

Libraries & Frameworks: NumPy, Pandas, Matplotlib

Tools & Technologies: MySQL, Docker, Qdrant

Collaboration & Communication: Stakeholder Engagement, Cross-Functional Collaboration, Leadership with ownership, Business Communication, High-Pressure Decision-Making, Structured Problem Solving

POSITIONS OF RESPONSIBILITY

Core, Production & Pronites -- Elan & nVision

- Managed end-to-end production logistics for a 10,000+ attendee techno-cultural fest, coordinating multiple teams under tight deadlines and high operational pressure

- Served as primary point of contact for headlining artists and bands, handling contracts, riders, and on-ground logistics for high-profile external partners
- Delivered flawless execution of large-scale pronite events, translating meticulous planning into a seamless attendee experience

Coordinator, International Relations Cell -- Inbound Domain

- Served as primary point of contact for incoming exchange students from Japan, managing onboarding and orientation from arrival through settlement
- Facilitated clear communication across language and cultural differences, contributing to a consistently smooth exchange program experience

House Captain - Mithra House, Meridian School

RELEVANT COURSES

Mathematics: Probability and Random Variables, Introduction to Statistics, Linear Algebra, Calculus, Discrete Mathematics, Transform Techniques, Complex Variables, Numerical Methods

Computer Science: Introduction to Programming, Data Structures, Algorithms, Operating Systems, Theory of Computation

EXTRACURRICULAR

Delegate, OakMUN, Chirec MUN & Meridian MUN (2018 and 2019) : Built and defended data-backed policy positions through structured negotiation, driving consensus among diverse, cross-cultural stakeholders on complex global issues.

Founder & Organizer, Peer Tutoring Club: Initiated and led a student-run tutoring program in Math and Science, coordinating peers to deliver regular academic support to junior students.

Basketball (Inter IIT Team): College Team in first and second year

Kung-Fu: Maroon Belt (second degree) - Won in several state level competitions during school.